

## Course Content- Advance Engineering Mathematics I

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### Unit 0: Motivation

**1 Lecture Hour**

Why study this course? Application of Calculus and Differential equations in addressing real-world problems.

### Unit I: Differential Calculus

**9 Lecture Hours**

Review: Functions and their graphs, polynomial, exponential, and logarithmic functions, Remainder and Factor theorems of polynomials.

Limits, Continuity and Differentiability, Rolle's theorem, Lagrange's and Cauchy mean value theorems, Successive differentiation, Leibnitz's theorem, Taylor's series without proof, Lagrange's form of remainder, Functions of several variables, Partial differentiation, Euler's theorem, Jacobian, Maxima and minima. Recap of Unit-I.

### Unit II: Integral Calculus

**9 Lecture Hours**

Definite integrals and properties, Double integrals: Cartesian and Polar co-ordinates, Cylindrical and Spherical coordinates, Change of order of integration, Change of variables, Triple integrals, Gamma, Beta functions and their properties, Applications of Calculus: Real-world problems. Recap of Unit-II.

### Unit III: Vector Algebra and Calculus

**9 Lecture Hours**

Motivation, Vector algebra, Scalar and vector point functions, Vector Differentiation, Gradient, Divergence and Curl, Vector Integration: Line integral, Surface integral, Volume integral, Application of integrals: Work, Circulation and Flux, Green, Gauss, and Stokes theorems (without proof) and their applications. Recap of Unit-III.

### Unit IV: Linear Differential Equations with Visualizations

**11 Lecture Hours**

Motivation: Linearity vs non-linearity, Linear superposition principle, First-order linear differential equations: Exact differential equations. Integrating Factors. Applications of first-order linear equations: Electric circuits, Radioactive decay, Population growth. Second and Higher order linear differential equations: Solution of homogeneous and non-homogeneous equations with constant coefficients, Wronskian, Solution of second-order differential equation by variation of parameters, Autonomous vs Non-autonomous systems, Applications: Harmonic oscillator, Electric Circuits. Visualization Tool: GeoGebra <https://www.geogebra.org/t/differential-equation>. Recap of Unit-IV.

### Unit V: Non-Linear Differential Equations with Visualizations

**6 Lecture Hours**

Brief history of interdisciplinary studies of nonlinear systems, Pendulum equations, Phase portraits, Linearization around equilibrium, Nonlinear models: Logistic equation, Bass model of diffusion of innovation, Lotka-Volterra predator-prey model, Epidemic models-SIR, Application to the nerve impulse. Visualization Tool: GeoGebra <https://www.geogebra.org/t/differential-equation>. Recap of Unit V.

### Text Books:

1. Thomas, G. B. and Finney, R. L., Calculus and Analytical Geometry, Pearson Education India, ISBN: 9788174906168, 2010.
2. James, G. and Dyke, P. P., Advanced Modern Engineering Mathematics. Pearson Education. ISBN: 9781292174341, 2018.

3. Kreyszig E., Advanced Engineering Mathematics, Wiley Publications. ISBN: 9780471488859, 2005.

#### **Reference Books:**

1. Apostol T. M., Mathematical Analysis, Narosa. ISBN: 9788185015668, 2002.
2. Braun M., Differential Equations and their Applications, Springer, ISBN: 9780387943305, 1994.
3. Strogatz S. H., Nonlinear Dynamics and Chaos, CRC Press, ISBN: 9781138329454, 2014.
4. Enns R. H., It's a Nonlinear World, Springer, ISBN: 978-0387753386, 2011.

#### **Web Resources and Supplementary Materials:**

1. Strogatz, S. Love Affairs and Differential Equations. 1988.  
<https://userpages.umbc.edu/~rostantia/math481/love-dynamics/love-dynamics-1988.pdf>
2. <https://blogs.cornell.edu/info2040/2018/11/29/sir-model-epidemics-and-social-media/>
3. <https://www.jstor.org/stable/1252012>.
4. Visualization Tool: GeoGebra <https://www.geogebra.org/t/differential-equation>.
5. Givon, M., Mahajan, V. and Muller, E.. Software piracy: Estimation of lost sales and the impact on software diffusion. Journal of Marketing, 59(1), 29-37,1995  
<https://www.jstor.org/stable/1252012>.
6. Singh, A., Basic Calculus. <https://archive.nptel.ac.in/courses/111/106/111106146/>
7. MIT OCW on Differential Equations.  
[https://ocw.mit.edu/courses/18-03-differential-equations-spring-2010/video\\_galleries/video-lectures/](https://ocw.mit.edu/courses/18-03-differential-equations-spring-2010/video_galleries/video-lectures/)

#### **Suggested Scope for Future Enquiry:**

Students may like to explore

- i) Maxwell's equations in electrodynamics,
- ii) Formulation of differential equations for Double pendulum.